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Development and validation of a burnout scale for undergraduate students in English as a medium of instruction programs

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Burnout is a growing concern for students in English as a Medium of Instruction (EMI) programs, where academic demands combine with language-related challenges. This study developed and validated a 17-item EMI Burnout Scale to address the lack of a context-specific measurement tool. The scale measures four related dimensions: Exhaustion, Cynicism, Disengagement, and Academic Efficacy. Items were adapted from the Maslach Burnout Inventory–General Survey for Students and the Oldenburg Burnout Inventory, with modifications for EMI contexts. Data were collected from 497 undergraduates in Saudi universities, representing science, engineering, healthcare, business, and computing disciplines. Partial Least Squares Structural Equation Modeling with a Reflective–Reflective Higher-Order Construct model was applied to test the scale. Correlation analysis showed strong positive relationships among Exhaustion, Cynicism, and Disengagement. Academic Efficacy was negatively associated with all three, indicating a potential protective role. The measurement model demonstrated high internal consistency, strong convergent validity, and clear discriminant validity, supported by AVE values and item loading patterns. These results confirmed that each dimension was measured accurately and without substantial overlap. Structural model evaluation indicated that Exhaustion, Cynicism, and Disengagement contributed positively to the higher-order burnout construct, while Academic Efficacy contributed negatively. Exhaustion emerged as the most influential factor and showed the highest predictive relevance. Predictive tests also confirmed that Exhaustion had the strongest forecasting ability, followed by Cynicism and Disengagement. The findings highlight the role of emotional strain in EMI burnout, shaped by the cognitive load of learning in a second language. The EMI Burnout Scale provides a valid and reliable tool for identifying students at risk. It can support targeted interventions such as language assistance, inclusive teaching strategies, and mental health resources. The study adds a context-specific instrument to burnout research and supports the development of sustainable EMI learning environments that address both academic and emotional needs.

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Introduction

Burnout is defined by the World Health Organization (2019) under International Classification of Diseases 11th code QD85, as a syndrome caused by unmanaged chronic workplace stress. It is marked by three main components: exhaustion, cynicism or mental distance from work, and reduced effectiveness. Although first linked to the workplace, burnout also occurs in other high-pressure settings, including academic environments. Schaufeli et al. (2002) extended this concept to students, describing it as emotional exhaustion, detachment from study tasks, and feelings of inadequacy. These academic pressures become even more complex in contexts where English is the language of instruction.

English as a medium of instruction (EMI) refers to teaching academic subjects in English in settings where English is not the primary language (Macaro, 2018). While EMI is often introduced to improve global competitiveness, it can also disadvantage students. Learners with lower English proficiency may experience cognitive overload, difficulty understanding lectures, and challenges in retaining content (Alhamami 2023a; Charoenpornsook and Thumvichit, 2025; Li and Pei, 2024). EMI is sometimes hindered by unclear communication, lack of language support, and ineffective teaching strategies. These limitations can lead to weaker academic outcomes and reduced classroom interaction (Espinas and Fuchs, 2022), increasing the strain that may contribute to burnout. By contrast, instruction in a student's native language often supports better comprehension, active participation, and academic confidence (Hennebry and Gao, 2021; Thompson et al., 2022).

Burnout in EMI settings can also be intensified by structural inequalities. These include insufficient English preparation in secondary education, weak institutional support, and limited instructor awareness of linguistic needs. In technical fields such as healthcare and engineering, students often face complex material in English without adequate preparation (Gaffas, 2025; Hamad et al., 2025). Instructors may use advanced English without clarification, while students are left to translate and interpret materials independently. The lack of learning resources in the native language can further limit understanding. Together, these academic and linguistic demands create an environment that increases vulnerability to burnout (Alhamami 2023b).

Research shows that this vulnerability is closely linked to psychological factors. Studies have connected academic burnout to anxiety, self-efficacy, and expectations for the future. Lizarte Simón et al. (2024) found that self-efficacy and psychological well-being can reduce the negative effect of anxiety on engagement. Hong (2025) reported that burnout plays a stronger mediating role than engagement between students' future expectations and academic performance. Xu et al. (2025) showed that willingness to communicate in EMI settings depends on emotional states, with enjoyment increasing participation and anxiety reducing it. Such findings suggest that improving students' emotional well-being could be a valuable strategy for lowering burnout in EMI contexts.

Despite these insights, burnout in EMI programs remains under-researched. While many studies have explored EMI challenges, few have directly measured burnout in these environments. Currently, no validated scale exists to assess the specific burnout experiences of EMI students. This study addresses that gap by developing and validating a burnout scale designed for EMI undergraduate programs. The instrument aims to provide context-specific data that can guide strategies to improve student well-being and academic success.

Literature review

Understanding burnout as a construct is essential for examining its occurrence in academic settings, including EMI contexts. The concept was first introduced in the 1970s by Herbert Freudenberger to describe emotional exhaustion among healthcare professionals. Christina Maslach later expanded the definition and developed the Maslach Burnout Inventory (MBI), which remains one of the most widely used tools for measuring burnout (Maslach et al., 2017; Schaufeli, 2017). Although initially associated with human services professions, burnout has since been applied in higher education. Students engaged in structured, goal-oriented activities such as lectures, assignments, and examinations may experience burnout in ways similar to working professionals (Hu and Schaufeli, 2009; Li et al., 2024).

Burnout is typically described through three dimensions: emotional exhaustion, cynicism, and inefficacy (Portoghese et al., 2018). Emotional exhaustion results from prolonged stress, cynicism reflects detachment or negative attitudes toward work, and inefficacy refers to a reduced sense of competence. These dimensions are evident in academic contexts, where heavy workloads and high expectations can lead to fatigue, disengagement, and weaker performance (Li et al., 2024). Several instruments have been developed to measure burnout. The MBI remains the most widely used, assessing emotional exhaustion, cynicism, and professional efficacy (Maslach and Leiter, 2016). The Oldenburg Burnout Inventory (OLBI) is also widely applied, focusing on exhaustion and disengagement across professions (Demerouti et al., 2010; Sinval et al., 2019). Other notable tools include the Copenhagen Burnout Inventory (Kristensen et al., 2005) and the Shirom–Melamed Burnout Measure (Shirom and Melamed, 2006). However, these instruments are not designed to capture the unique pressures found in EMI programs, highlighting a gap in existing measurement tools.

Insights from applied linguistics research provide further understanding of how burnout develops in language learning environments. In recent years, studies have increasingly examined burnout in English as a foreign language (EFL) contexts. Although the number of studies remains limited, research has investigated links between burnout and psychological traits, teacher–student relationships, and emotional regulation. Fan et al. (2024) reported that growth mindset and grit may reduce burnout among Chinese high school students. Kashirskaya et al. (2024) linked burnout with creativity and multilingualism in university students. Teacher–student interaction also matters: Wu et al. (2023) and Derakhshan et al. (2022) found that affective teacher support may lower burnout risk. These insights are relevant to EMI programs where language and content demands occur together.

Emotional regulation appears to play a central role in burnout outcomes for language learners. Wang et al. (2024) used network analysis to show that maladaptive strategies, such as avoidance, can intensify burnout. Gao (2024) examined mindfulness as a protective factor, while Li et al. (2024) designed tools to assess burnout in EFL contexts. Ding and Wang (2024) conducted a longitudinal study with 621 Chinese EFL students, finding that although burnout decreased over time, negative emotions—including anxiety, boredom, and hopelessness—remained strong predictors. These results suggest that burnout in language learning is shaped by multiple cognitive and emotional factors. They also indicate that promoting emotional well-being should be an important part of language education. This evidence is directly relevant to EMI programs, which combine language learning with content mastery, potentially intensifying burnout risk.

Research in EMI contexts has identified additional structural and linguistic challenges that can contribute to burnout. Block (2022) describes a "dark side" of EMI, highlighting potential inequalities for both students and instructors. Alhamami (2023b) reported that EMI policies often fail to ensure fairness, creating unequal conditions, especially for students with lower English proficiency. In many non-Anglophone settings, EMI is implemented without fully considering learners' linguistic needs. Sah and Li (2018) argue that such policies reinforce educational inequality, disproportionately affecting disadvantaged students. In Nepal, Sah and Karki (2020) documented a "comprehension crisis" among marginalized EMI students, which further deepened epistemic inequalities.

The language demands of EMI can directly increase burnout risk. Rose et al. (2020) found that Japanese students with lower proficiency struggled to meet academic expectations, while Bhattacharya (2013) reported similar challenges in India, where reliance on rote memorization was common due to limited comprehension. Such barriers can lower academic performance, reduce engagement, and heighten emotional stress (Siegel, 2020). In China, Dong and Han (2025) observed that international EMI students often faced isolation, anxiety, and emotional strain linked to language difficulties, academic workload, and social exclusion. Although coping strategies such as digital tools and peer support were used, they did not fully prevent emotional costs resembling academic burnout.

Meta-analytic research reinforces these patterns. Lee et al. (2025) found that English proficiency and engagement are critical to success in EMI higher education. Language anxiety and motivation strongly influence proficiency, which in turn affects academic achievement. These findings highlight the combined cognitive and emotional demands of EMI and emphasize the need for reliable, context-specific tools to assess burnout. In Saudi Arabia, Alhamami (2023b) noted that many students preferred instruction in their native language, reflecting negative attitudes toward EMI.

Existing burnout instruments, while validated in various professional and educational contexts, are not tailored to the combined linguistic, cognitive, and socio-cultural demands of EMI undergraduate programs. Studies in applied linguistics and EMI highlight the influence of language proficiency, emotional well-being, and institutional factors on burnout, yet no measurement tool addresses these variables in an integrated manner. This gap limits the ability to assess burnout accurately and develop targeted interventions for EMI students. The present study responds to this need by developing and validating a burnout scale specifically designed for EMI contexts, providing a foundation for future research and practical support strategies.

Study contributions and research questions

The present study directly addresses this gap by focusing on burnout in EMI undergraduate programs. While previous research has documented the presence of burnout in such environments (Alhamami, 2023a, 2023b; Block, 2022; Sah and Li, 2018; Sah and Karki, 2020), no validated tool has been developed to measure it in a way that reflects both the academic and linguistic realities of EMI learners.

This research contributes by designing and validating a 17-item burnout scale that incorporates the specific challenges faced by EMI students. The scale captures four related but distinct dimensions: Exhaustion, Cynicism, Disengagement, and Academic Efficacy.

A further methodological contribution is the application of Partial Least Squares Structural Equation Modeling (PLS-SEM) within a Reflective–Reflective Higher-Order Construct (R-R HOC) framework. Both the higher-order construct (burnout) and the lower-order constructs (the four dimensions) were modeled reflectively. This approach, seldom applied in academic burnout research (Cox et al., 2005; Qiao and Schaufeli, 2011), enables the assessment of multidimensional constructs without separate exploratory and confirmatory phases. The disjoint two-stage procedure, as recommended by Becker et al. (2023) and Hair et al. (2024), was used to estimate path coefficients between lower-order constructs (LOCs) and the higher-order construct (HOC), ensuring robust measurement and structural evaluation.

The measurement model examines internal consistency, convergent validity, and discriminant validity. The structural model assesses the strength and direction of relationships between dimensions, as well as the model's predictive accuracy. By providing a psychometrically sound, context-specific tool, this study advances EMI research and offers a practical method for identifying students at risk of burnout. The results can guide interventions such as targeted language support, student-centered pedagogy, and mental health services.

Research questions

RQ1: Correlation matrix. What are the relationships between Academic Efficacy, Exhaustion, Cynicism, and Disengagement among undergraduate students in EMI programs?

RQ2: Measurement model evaluation. To what extent do the four burnout LOCs—Academic Efficacy, Cynicism, Disengagement, and Exhaustion—demonstrate acceptable measurement properties (i.e., convergent validity, discriminant validity, and internal consistency reliability) within an R-R HOC model using PLS-SEM?

RQ3: Structural model evaluation. To what extent do the four burnout LOCs demonstrate structural significance, predictive relevance, and inter-construct correlations within the R-R HOC model?

Methodology

This study employed a quantitative, cross-sectional survey design to develop and validate a burnout scale tailored to EMI undergraduate programs. The methodology was structured to ensure both conceptual relevance and statistical rigor. The process involved adapting items from established burnout instruments, translating and reviewing the scale for contextual suitability, and collecting data from a diverse sample of EMI students in Saudi universities. The analysis applied PLS-SEM within a R-R HOC framework to assess the measurement and structural properties of the scale. The following subsections describe the scale design, study procedures, and data analysis in detail.

Scale design

This study measured burnout in EMI undergraduate students using items adapted from two established instruments: MBI and OLBI. The MBI is widely applied to assess burnout across professional and educational contexts, with versions adapted for human services, healthcare, education, and student populations (Maslach et al., 2017; Portoghese et al., 2018). For this study, the Maslach Burnout Inventory – General Survey for Students (MBI-GS[S]) was used. It contains three constructs: Exhaustion, Cynicism, and Academic Efficacy (Maslach et al., 2017).

The OLBI was included to capture Disengagement, a dimension not fully addressed by the MBI. The OLBI consists of 16 items—eight on Exhaustion and eight on Disengagement (Demerouti et al., 2010). Since Exhaustion was already measured by the MBI, only Disengagement items were taken from the OLBI. The final 17-item scale therefore, included five Exhaustion items, four Cynicism items, and four Academic Efficacy items from the MBI, along with four Disengagement items from the OLBI. This combination was designed to reflect burnout experiences specific to EMI learning contexts.

We anchored every item to the EMI context. We used the clause "due to the use of English as the language of instruction" in all items, either directly or with similar wording. This fixes the cause of stress or attitudes to the instructional medium rather than to general study demands. We also aligned domains by changing OLBI "work/job" to "studies/academic tasks" and by focusing MBI "studies" to "my major/classes." We replaced idiomatic or culture-bound phrases with plain, L2-friendly wording to support translation and comprehension (e.g., "used up" → "exhausted," "getting things done" → "accomplishing tasks," "a strain" → "stressful"). For Academic Efficacy, we retained capability and contribution items and omitted broad self-concept statements to keep the construct task-focused in EMI settings. From the OLBI we kept disengagement indicators and did not include reverse-keyed engagement items to avoid mixed valence. These edits preserve the original constructs while reflecting EMI-specific experiences. Our reliability and validity results support this claim.

Academic Efficacy was treated as a construct opposite to the other three burnout dimensions. Reversed items were used for this construct to improve measurement validity in several ways. First, they help reduce acquiescence bias, where respondents tend to agree with statements regardless of content (Baumgartner and Steenkamp, 2001). Second, reversed items interrupt automatic response patterns, encouraging respondents to consider each answer carefully (Podsakoff et al., 2003). Third, they allow the measurement of both positive and negative aspects of the construct, providing broader coverage (Tourangeau et al., 2000). While reversed items may sometimes lower reliability or produce inconsistent responses, their inclusion supports balanced measurement and helps capture both protective and risk factors relevant to EMI burnout. The complete list of items is provided in the appendix.

Study procedures

The research process included ethical approval, instrument adaptation, translation, expert review, and data collection. Ethical approval was granted by the Institutional Review Board at King Khalid University, and the study followed the Research Ethics Committee's guidelines and the Declaration of Helsinki. Participants were informed about the study's aims, assured of voluntary participation and confidentiality, and gave informed consent before completing the survey. All data were stored securely.

The online survey consisted of three sections. The first gathered demographic data such as gender, major, and current year of study. The second included the 17 burnout items adapted from the MBI-GS(S) and OLBI. The MBI-GS(S) items measured Exhaustion (5 items), Cynicism (4 items), and Academic Efficacy (4 items) (Maslach et al., 2017; Schaufeli et al., 2009). The OLBI contributed four items on Disengagement (Demerouti et al., 2010). All items were adapted to reflect EMI-specific academic experiences. The third section contained open-ended questions; these responses will be analysed in a separate qualitative study to maintain this paper's focus on quantitative scale validation.

A multi-step adaptation process was applied to ensure contextual suitability. The finalized English version was translated into Arabic to enhance clarity and reduce potential language barriers. Translation was carried out by the researchers, who are native Arabic speakers fluent in English and familiar with the academic environment. Four university instructors then reviewed both the English and Arabic versions to assess content validity, ensuring each item reflected its intended construct. Revisions were made to improve clarity, accuracy, and construct alignment. The final instrument used a six-point Likert scale from "strongly agree" to "strongly disagree" and was distributed via Google Forms.

The tool was designed for use in EMI programs across different cultural settings. It targeted students in science, engineering, and medical disciplines taught in English. The survey was distributed using multiple channels, including official university email lists, with faculty asked to forward it to eligible students. It was also shared via WhatsApp and Telegram, platforms widely used by Saudi university students. This approach increased sample size and diversity, reaching students from physics, engineering, pharmacy, and related programs.

Analysis procedures

The initial step in analysis was to screen the 523 completed survey responses. Only students currently enrolled in EMI programs were included, as the study aimed to capture the experiences of those with substantial EMI exposure. Twenty-six responses were excluded because participants were in preparatory-year programs or studying in fields taught mainly in Arabic, such as social sciences and Sharia law. Although preparatory-year students may later join EMI majors, their current studies focus mainly on English language courses rather than subject-specific EMI learning. This exclusion produced a final sample of 497 students with genuine EMI experience.

Responses were coded numerically, with Likert scale options assigned values from 6 ("strongly agree") to 1 ("strongly disagree"). Higher scores indicated greater burnout for Exhaustion, Cynicism, and Disengagement. For Academic Efficacy, coding was reversed so that higher scores indicated lower burnout. This ensured consistent interpretation across constructs.

PLS-SEM was used to examine measurement and structural relationships. The analysis applied a R-R HOC model in which burnout served as the HOC, composed of four LOCs: Exhaustion, Cynicism, Disengagement, and Academic Efficacy. Each LOC was measured by multiple related items. This hierarchical design enabled both an overall burnout score and insights into how each dimension contributed to the construct (Marôco and Campos, 2012). Prior research supports modeling burnout as a second-order factor, with exhaustion and disengagement often most strongly associated, and cynicism and reduced efficacy contributing to a lesser extent (Kristensen et al., 2005; Halbesleben and Demerouti, 2005; Schaufeli et al., 2009; Cox et al., 2005; Qiao and Schaufeli, 2011).

Descriptive statistics were generated in Jamovi to summarize demographic characteristics and mean scores for each burnout dimension. These preliminary results offered an overview of burnout patterns in the sample. PLS-SEM analysis was then conducted in SmartPLS using a two-stage approach. The first stage assessed the measurement model by examining indicator loadings, Cronbach's alpha, composite reliability, and convergent validity through average variance extracted (AVE). Discriminant validity was also tested to confirm that each LOC measured a distinct construct.

The second stage evaluated the structural model. Path coefficients were examined to identify relationships among constructs, R^2 values to determine explained variance, and effect sizes (f^2) to assess relationship strength. Predictive relevance was tested using Q^2 predict, alongside root mean square error (RMSE) and mean absolute error (MAE), following the guidelines of Ringle et al. (2024). All analyses were conducted to ensure the scale demonstrated both reliability and validity within the EMI context.

Results

This section presents the statistical findings from the analysis of the EMI Burnout Scale. It begins with a description of the sample

Table 1 Descriptive statistics of participants demographic information.

Variables	Levels	Count	Proportion
Gender	Female	307	0.618
	Male	190	0.382
Colleges	Business Colleges	76	0.153
	Computer Colleges	111	0.223
	Engineering Colleges	97	0.195
	Healthcare Colleges	145	0.292
	Sciences Colleges	68	0.137
Years of Study	1	202	0.406
	2	129	0.260
	3	71	0.143
	4	51	0.103
	5	29	0.058
	More than 5	15	0.030

demographics, followed by the results for each research question. These include correlation patterns among burnout dimensions, measurement model evaluation, and structural model testing. Additional analyses assess the predictive ability of the model.

Participants' descriptive statistics. The final sample included 497 EMI undergraduate students. Of these, 61.8% were female ($n = 307$) and 38.2% were male ($n = 190$). Participants represented diverse academic disciplines. The largest group was from Healthcare Colleges (29.2%), followed by Computer Colleges (22.3%), Engineering Colleges (19.5%), Business Colleges (15.3%), and Sciences Colleges (13.7%). Regarding academic level, 40.6% were in their first year after the Preparatory Year Program, 26.0% were in the second year, 14.3% in the third year, 10.3% in the fourth year, 5.8% in the fifth year, and 3.0% had studied for more than five years. Students were enrolled in multiple universities. The largest representation came from Qassim University (11.9%), King Khalid University (10.5%), and King Saud University (9.1%). Other universities with notable participation included Najran University (7.8%), Imam Mohammad Ibn Saud Islamic University (7.2%), and King Fahd University of Petroleum and Minerals (7.0%). Several other institutions contributed smaller proportions of participants. Table 1 presents detailed demographic information.

RQ1: Correlation matrix. This subsection examines the relationships among the four LOCs of burnout. Mean scores, standard deviations, and correlation coefficients are presented to show the strength and direction of these relationships.

Before the correlation analysis in Jamovi, mean scores were calculated for each LOC of burnout. All assumptions for correlation analysis were tested and met.

Scores ranged from 1 to 6. For Exhaustion, Cynicism, and Disengagement, higher scores indicated greater burnout. For Academic Efficacy, higher scores indicated lower burnout due to the reverse nature of this construct. The mean for Academic Efficacy was 3.22 (SD = 1.59), suggesting moderate perceived efficacy. Higher mean scores were observed for Exhaustion (M = 4.31, SD = 1.73), Disengagement (M = 4.04, SD = 1.73), and Cynicism (M = 3.78, SD = 1.89).

The correlation analysis reveals significant relationships between the four constructs. Exhaustion is strongly correlated with both Cynicism ($r = 0.869$, $p < 0.001$) and Disengagement ($r = 0.866$, $p < 0.001$). Cynicism and Disengagement are also highly correlated ($r = 0.863$, $p < 0.001$). On the other hand, Academic Efficacy is negatively correlated with Exhaustion ($r = -0.571$, $p < 0.001$), Cynicism ($r = -0.543$, $p < 0.001$), and Disengagement ($r = -0.520$, $p < 0.001$), suggesting that higher efficacy is associated with lower burnout levels. Table 2 presents the means, standard deviations, and correlations for the LOCs of burnout.

Figure 1 illustrates the pairwise correlations between the means of the four constructs of burnout: M-Academic Efficacy, M-Cynicism, M-Disengagement, and M-Exhaustion. The correlation coefficients (r) are displayed within the plot, where positive values indicate direct relationships, and negative values represent inverse relationships between the constructs. The statistical significance of each correlation is indicated by asterisks, with $^{***}p < 0.001$.

RQ2: Measurement model evaluation. This subsection evaluates the measurement properties of the EMI Burnout Scale. Results for convergent validity, reliability, and discriminant validity are reported to confirm the quality of the constructs.

Convergent validity. The convergent validity analysis for both the LOCs and the HOC showed strong results. AVE values exceeded the recommended threshold of 0.50 for all LOCs—Academic Efficacy (0.81), Cynicism (0.87), Disengagement (0.81), and Exhaustion (0.88). The HOC, EMI Burnout, also achieved an AVE of 0.674. These results indicate that the constructs capture a high proportion of variance from their indicators, satisfying convergent validity requirements (Fornell and Larcker, 1981). Table 3 presents the detailed convergent validity results for all constructs.

Reliability. All LOCs demonstrated excellent internal consistency. Cronbach's alpha values ranged from 0.92 to 0.97, exceeding the 0.70 benchmark. Composite reliability values (rho_a and rho_c) were also well above the recommended level of 0.70, confirming reliability: Academic Efficacy (0.92, 0.94), Cynicism (0.95, 0.96), Disengagement (0.92, 0.94), and Exhaustion (0.97, 0.97). The HOC also showed strong reliability with Cronbach's alpha of 0.877 and composite reliability values of 0.973 (rho_a) and 0.933 (rho_c). Table 3 presents the detailed reliability results for all constructs.

Discriminant validity. The discriminant validity of the EMI Burnout Scale was supported by the item loadings and cross-loadings. Each LOC showed strong outer loadings on its own construct and weaker loadings on other constructs. Academic Efficacy items loaded between 0.869 and 0.924 on their construct, with lower correlations to Cynicism and Disengagement. Cynicism items loaded from 0.894 to 0.956, Disengagement from 0.906 to 0.944, and Exhaustion from 0.930 to 0.950. This pattern confirms that each construct was measured uniquely without substantial overlap (Table 4).

Further assessment using the Fornell–Larcker Criterion confirmed discriminant validity (Hair et al., 2019; Sarstedt et al., 2014). The square root of AVE for each construct—Academic Efficacy (0.898), Cynicism (0.932), Disengagement (0.899), and Exhaustion (0.940)—exceeded the correlations with other constructs. For example, Academic Efficacy had higher AVE (0.898) than its correlations with Cynicism (0.545), Disengagement (0.524), and Exhaustion (0.572). Similar patterns were observed for the other LOCs, confirming that all constructs are statistically distinct within the model.

RQ3: Structural model evaluation. This subsection assesses the structural relationships between the higher-order burnout construct and its dimensions. It also reports the model's predictive ability using Q^2predict, RMSE, and MAE values.

Table 2 Correlation matrix of burnout LOCs.

LOCs	M(SD)	M-Academic Efficacy	M-Cynicism	M-Disengagement	M-Exhaustion
M-Academic Efficacy	3.22(1.59)	—			
M-Cynicism	3.78(1.89)	−0.543***	—		
M-Disengagement	4.04(1.73)	−0.520***	0.863***	—	
M-Exhaustion	4.31(1.73)	−0.571***	0.869***	0.866***	—

*p < 0.05, **p < 0.01, ***p < 0.001.

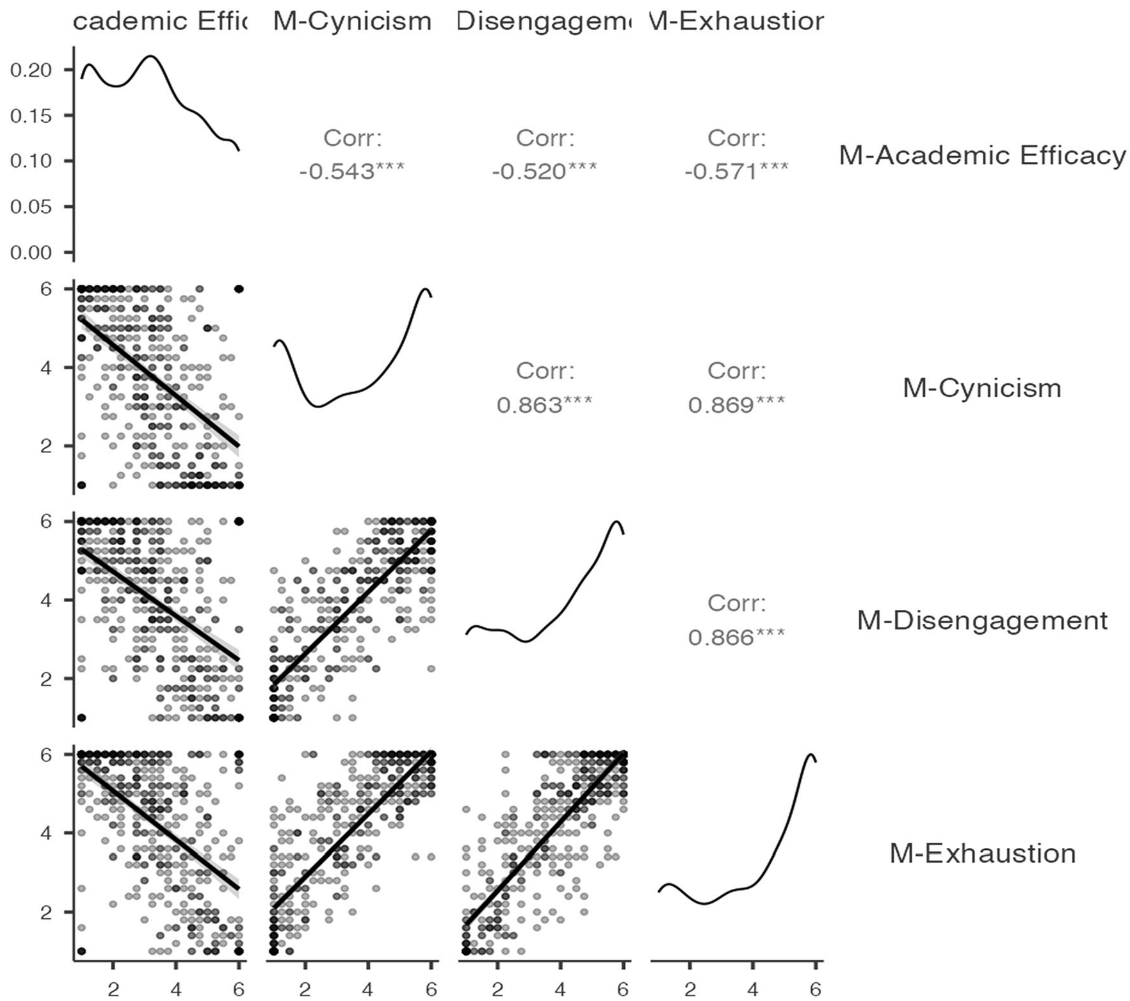

Fig. 1 Correlation Matrix Plot of Burnout LOCs.

Table 3 Convergent validity and reliability.

Construct	Cronbach's Alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	AVE
LOC: Academic Efficacy	0.92	0.92	0.94	0.81
LOC: Cynicism	0.95	0.95	0.96	0.87
LOC: Disengagement	0.92	0.92	0.94	0.81
LOC: Exhaustion	0.97	0.97	0.97	0.88
HOC: EMI Burnout	0.877	0.973	0.933	0.674

Structural contribution. The results of the structural model evaluation presented in Fig. 2 and Table 5 demonstrate the relationships between the HOC of EMI Burnout and its associated LOCs: Academic Efficacy, Cynicism, Disengagement, and Exhaustion. The model reveals strong, statistically significant path coefficients between EMI Burnout and each of its LOCs. Notably, Exhaustion exhibits the highest path coefficient (0.955), reflecting a strong positive relationship, with an R^2 value of 0.912, indicating that this LOC explains a significant portion of the variance in EMI Burnout. Similarly, Cynicism (0.938) and Disengagement

Table 4 Discriminant validity of the burnout EMI scale.

Items	Academic Efficacy	Cynicism	Disengagement	Exhaustion	EMI Burnout
Academic Efficacy1	0.869	−0.446	−0.429	−0.476	−0.588
Academic Efficacy2	0.924	−0.472	−0.47	−0.504	−0.627
Academic Efficacy3	0.893	−0.525	−0.478	−0.523	−0.644
Academic Efficacy4	0.905	−0.513	−0.5	−0.548	−0.658
Cynicism1	−0.526	0.945	0.82	0.834	0.894
Cynicism2	−0.549	0.956	0.862	0.864	0.923
Cynicism3	−0.5	0.932	0.798	0.807	0.869
Cynicism4	−0.452	0.894	0.74	0.732	0.806
Disengagement1	−0.496	0.797	0.906	0.804	0.857
Disengagement2	−0.369	0.716	0.864	0.692	0.757
Disengagement3	−0.477	0.766	0.909	0.781	0.837
Disengagement4	−0.528	0.826	0.914	0.835	0.885
Exhaustion1	−0.551	0.785	0.796	0.93	0.882
Exhaustion2	−0.539	0.823	0.833	0.949	0.907
Exhaustion3	−0.517	0.818	0.803	0.927	0.885
Exhaustion4	−0.529	0.827	0.821	0.95	0.903
Exhaustion5	−0.552	0.837	0.828	0.944	0.91

Items highlighted in yellow represent the highest loadings for each respective construct, confirming their discriminant validity.

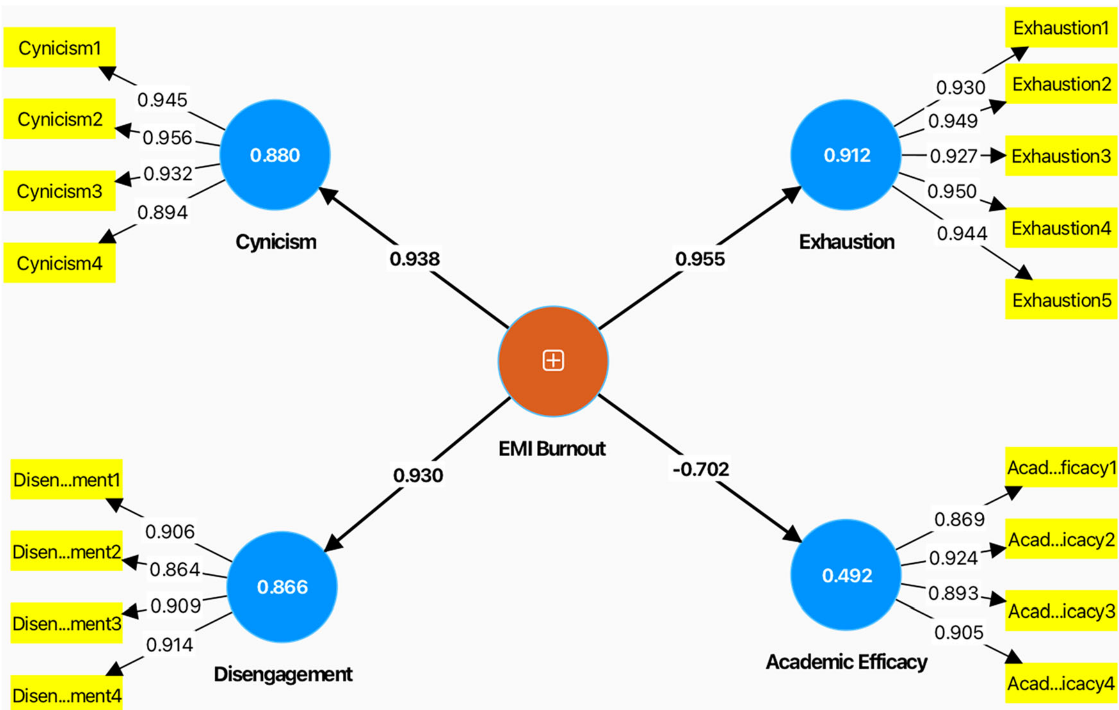

Fig. 2 EMI Burnout Scale*: Reflective Measurement Model.*

Table 5 Structural Model Evaluation.

HOC	LOCs	F2	R2	Path coefficient(O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (|O/STDEV|)	*P* value
EMI Burnout Scale	Academic Efficacy	0.970	0.492	−0.702	−0.702	0.036	19.527	0.000
	Cynicism	7.362	0.880	0.938	0.938	0.005	179.825	0.000
	Disengagement	6.441	0.866	0.930	0.930	0.007	139.847	0.000
	Exhaustion	10.334	0.912	0.955	0.955	0.004	240.004	0.000

(0.930) also show strong positive relationships with EMI Burnout, with high R^2 values of 0.880 and 0.866, respectively, further confirming the substantial role these constructs play in defining the burnout scale.

On the other hand, Academic Efficacy shows a strong negative path coefficient (−0.702), with an R^2 value of 0.492, signifying its inverse relationship with burnout. High academic efficacy correlates with lower levels of burnout, as the construct measures

Table 6 Cross-validated predictive ability test.

LOCs	Q^2predict	RMSE	MAE
Academic Efficacy	0.491	0.716	0.519
Cynicism	0.880	0.348	0.266
Disengagement	0.865	0.368	0.271
Exhaustion	0.912	0.299	0.233

the perceived lack of burnout. The effect sizes (f^2) for each LOC are notably high, particularly for Exhaustion (10.334) and Cynicism (7.362), emphasizing their critical contribution to the overall model. The T-statistics and p-values also confirm that these relationships are statistically significant, with values well above the threshold, ensuring the robustness of the structural model. Overall, the model highlights the pivotal role of exhaustion, cynicism, and disengagement in contributing to burnout, with academic efficacy acting as a mitigating factor.

Cross-validated predictive ability test. The Cross-Validated Predictive Ability Test assesses how effectively the model predicts burnout-related outcomes, using metrics such as Q^2predict, RMSE, and MAE. A Q^2predict value greater than zero indicates the model's predictive relevance, while lower RMSE and MAE values reflect better prediction accuracy. The results show that Exhaustion has the highest predictive relevance ($Q^2\text{predict} = 0.912$), with low error rates ($RMSE = 0.299$, $MAE = 0.233$), indicating strong predictive accuracy. Cynicism also demonstrates robust predictive relevance ($Q^2\text{predict} = 0.880$, $RMSE = 0.348$, $MAE = 0.266$). Disengagement follows closely with a Q^2predict value of 0.865 and low error rates ($RMSE = 0.368$, $MAE = 0.271$). However, Academic Efficacy has a lower Q^2predict value of 0.491 and higher error rates ($RMSE = 0.716$, $MAE = 0.519$), suggesting moderate predictive relevance. Table 6 summarizes the Cross-Validated Predictive Ability Test results.

Discussion

This study addressed a notable gap in the literature by developing and validating a burnout scale tailored to EMI undergraduate programs. While existing instruments measure burnout in general academic or occupational contexts, they often overlook the combined impact of academic demands and language-related challenges in EMI settings. The scale presented here captures four interrelated but distinct dimensions: exhaustion, cynicism, disengagement, and academic efficacy. These dimensions reflect the layered pressures that students face when learning through a non-native language, particularly in contexts where high cognitive load meets sustained emotional and physical demands (Schaufeli et al., 2002; World Health Organization, 2019). Earlier research has indicated that these pressures may be intensified by structural inequities and limited institutional resources (Alhamami, 2023b; Hamad et al., 2025; Sah and Karki, 2020), making a targeted measurement tool essential.

The correlation analysis provides insight into how these dimensions interact. Exhaustion, cynicism, and disengagement show strong positive associations, suggesting that emotional fatigue is closely linked with negative attitudes and withdrawal from studies. In contrast, academic efficacy is negatively associated with all three, indicating that students with higher confidence in their academic abilities may be more resilient to burnout. These results are consistent with established burnout frameworks, which describe emotional depletion and detachment as reinforcing processes, while self-efficacy acts as a protective element. Linking these findings with previous research suggests that interventions focusing on strengthening academic efficacy could be particularly effective in EMI contexts.

The measurement model results confirm that the EMI Burnout Scale meets rigorous psychometric standards. All constructs achieved acceptable convergent validity, with substantial variance explained by their indicators. Reliability was high for both the LOCs and the HOC, and discriminant validity was supported by item loading patterns and the Fornell–Larcker criterion. These findings suggest that the scale measures burnout dimensions distinctly and consistently. This level of validity is crucial in EMI research, where measurement errors could obscure the specific effects of language and academic demands.

The structural model results further strengthen the case for the scale's usefulness. Exhaustion, cynicism, and disengagement contributed most positively to the HOC of burnout, while academic efficacy had a substantial negative contribution. Effect sizes confirmed the leading roles of exhaustion and cynicism, both in defining burnout and in predicting related outcomes. Although academic efficacy had a more modest predictive relevance, its protective effect remains important. This pattern suggests that reducing exhaustion and cynicism should be a short-term priority, while fostering academic efficacy can help maintain resilience over the long term.

The central role of exhaustion aligns with the view that sustained cognitive and linguistic demands deplete emotional and physical resources (Maslach et al., 2017). In EMI contexts, processing academic content in a second language may accelerate fatigue compared with studying in a first language (Alhamami, 2023a; Hennebry and Gao, 2021). This effect appears stronger among students with lower English proficiency, who face additional challenges in comprehension and retention (Li and Pei, 2024; Charoenpornsook and Thumvichit, 2025). These findings highlight the need for integrated interventions that combine language support with stress management to sustain academic engagement.

The strong connections between exhaustion, cynicism, and disengagement have also been observed in previous burnout research (Schaufeli et al., 2002; Portoghese et al., 2018). In EMI programs, cynicism may emerge as a coping strategy when students feel overwhelmed by persistent academic and linguistic obstacles. This detachment, described as the "dark side" of EMI (Block, 2022), reflects both academic frustration and emotional strain. Similar trends have been noted in Japanese EMI contexts, where students with lower proficiency reported higher disengagement (Rose et al., 2020). These parallels suggest that EMI burnout has both global and context-specific features, requiring interventions that address academic and emotional needs simultaneously (Bhattacharya, 2013).

Moderate levels of academic efficacy in this study indicate that some students maintain confidence in their abilities despite high exhaustion or disengagement. This self-belief appears to reduce susceptibility to severe burnout, echoing earlier findings that link perceived competence to lower emotional fatigue (Li et al., 2024). The negative correlations between efficacy and other burnout dimensions (Fan et al., 2024) confirm this role. Strategies such as study skills training, time management workshops, and targeted language support could strengthen academic efficacy, thereby enhancing resilience in the face of EMI-related challenges (Alhamami, 2023b; Li and Pei, 2024; Charoenpornsook and Thumvichit, 2025).

The reinforcing cycle between exhaustion, cynicism, and disengagement highlights the importance of integrated approaches. Fatigue can lead to cynicism, which in turn prompts disengagement, creating a downward spiral (Maslach et al., 2017). In EMI programs, this cycle may be accelerated by language-related frustration, particularly for students with limited proficiency (Sah and Karki, 2020). Breaking this cycle may require coordinated interventions that target emotional, academic, and linguistic factors at the same time, rather than addressing each in isolation.

The use of the R-R HOC model provides a valuable framework for understanding burnout as a set of related but distinct constructs (Cox et al., 2005; Qiao and Schaufeli, 2011). In EMI settings, where academic, linguistic, and emotional demands overlap, this approach offers a clearer picture of how different burnout components interact (Maslach and Leiter, 2016). Predictive analyses from this study underscore the importance of exhaustion as both a prevalent and predictive factor (Maslach et al., 2017). While academic efficacy had the lowest predictive power, its role in supporting long-term resilience justifies sustained attention (Fan et al., 2024).

Overall, these findings indicate that burnout in EMI undergraduate programs is shaped by the combined effects of emotional strain and academic demands. The validated multidimensional model shows exhaustion as central, cynicism and disengagement as emotional responses to persistent challenges, and academic efficacy as a moderating factor. Addressing these dimensions in an integrated manner appears essential for balancing academic performance with student well-being (Gaffas, 2025; Hamad et al., 2025).

Practical measures should target both language and academic support. Preparatory English courses, in-course assistance, and continuous proficiency workshops can help students manage linguistic demands. Regular monitoring of burnout using validated tools, such as the present scale, could allow institutions to identify issues early and provide timely interventions.

Emotional support should be embedded into EMI program structures. Counseling services, resilience-building activities, and stress management workshops can help reduce the connection between exhaustion and cynicism. At the classroom level, teachers can lower cognitive load by using visual aids, glossaries, and clear, paced explanations without reducing academic standards.

Active learning approaches can help prevent disengagement. Group projects, peer collaboration, and project-based learning can improve motivation and connect new material with students' prior experiences. Workshops on study strategies, time management, and self-directed learning can further strengthen academic efficacy and help students cope with the dual challenge of language and content learning.

Faculty training is a key component of these strategies. Professional development should focus on recognizing burnout symptoms, adopting inclusive teaching practices, and integrating language scaffolding into subject instruction. Flexible assessment formats—such as collaborative projects, oral presentations, or open-book exams—can reduce language-related barriers while maintaining academic rigor.

This study provides both theoretical and practical contributions. Theoretically, it expands burnout research by applying an R-R HOC framework in an EMI context, highlighting the interaction between cognitive, linguistic, and emotional demands. Practically, it offers a validated instrument for identifying at-risk students and guiding targeted interventions. Future research could examine how burnout patterns differ across disciplines, language proficiency levels, and cultural contexts, or track changes over time to assess the long-term effectiveness of interventions. Policymakers and academic leaders should consider integrating burnout monitoring into quality assurance processes, ensuring that EMI programs promote both learning outcomes and student well-being.

Limitations and directions for future research. This study has several limitations that point to clear avenues for future work. First, the EMI Burnout Scale was validated only within Saudi Arabia. Although the instrument showed strong psychometric properties in this context, its stability in other countries and cultural settings remains untested. Future research should examine its applicability across diverse EMI systems to establish cross-cultural validity. Second, the study did not assess measurement invariance across gender, academic discipline, or year of study, despite observed differences in burnout experiences between these groups. Testing invariance in future studies would clarify whether scores are truly comparable across subgroups. Similarly, the scale was administered only in English and Arabic; translations into other foreign languages should be developed and validated to broaden its accessibility.

Third, while exhaustion, cynicism, and disengagement emerged as dominant burnout dimensions, their links to concrete academic outcomes—such as GPA, credit completion, and program retention—were not explored here. Future research should examine these relationships to provide evidence for the predictive utility of the scale. Fourth, this study focused on undergraduates in general EMI contexts. Adapting and testing the instrument for other educational stages, including high school and postgraduate EMI programs, could determine whether similar burnout patterns occur across academic levels. Fifth, although academic efficacy appeared to play a protective role, the present design was cross-sectional. Pre- and post-intervention studies, using strategies such as language support, resilience training, or student-centered pedagogy, could assess the scale's sensitivity to change and evaluate intervention effectiveness. Finally, delivery mode was not considered as a variable. Comparing burnout across online, face-to-face, and blended EMI learning environments in future studies may reveal which formats reduce or exacerbate burnout, offering practical guidance for designing supportive EMI programs.

Conclusion

This study developed and validated a scale to measure burnout among EMI undergraduate students, addressing a clear gap in the literature. Burnout emerged across four interrelated constructs—exhaustion, cynicism, disengagement, and academic efficacy—reflecting the combined cognitive, linguistic, and emotional challenges in EMI contexts. Exhaustion, intensified by high academic demands and language barriers, was the most prominent dimension, while cynicism and disengagement also played substantial roles. Academic efficacy acted as a protective factor, though its moderate levels suggest it cannot fully counter burnout. Using an R-R HOC model allowed a comprehensive assessment of these dimensions, confirming their distinct yet connected nature, and demonstrating strong reliability, validity, and predictive capacity. The prominence of exhaustion, cynicism, and disengagement highlights the need for targeted strategies to reduce emotional strain, sustain engagement, and build academic efficacy. These findings emphasize the importance of integrated support systems in EMI programs, combining language assistance, student-centered teaching, and accessible psychological services to address both academic and emotional demands. By offering a reliable diagnostic tool and evidence-based guidance, this study contributes to the literature on higher education burnout and supports the creation of more sustainable and supportive EMI learning environments.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to privacy of the participants and the university databases but are available from the corresponding author on reasonable request.

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Author contributions

M.A. wrote the main manuscript and designed the research instrument. A.A. revised the manuscript, assisted in designing the instrument, and contributed to data collection.

Competing interests

The authors declare no competing interests.

Ethical approval

Ethical approval for this survey study was obtained from the Research Ethics Committee at King Khalid University (approval number ECM#2024-3125; approval date October 16, 2024; approval valid until October 15, 2025). All procedures involving the participants were conducted in accordance with the committee's regulations (HAPO-06-B-001) and the ethical principles of the Declaration of Helsinki and its later amendments or comparable ethical standards. The approval covered the protocol titled Academic Burnout in EMI Programs, including recruitment and online questionnaire data collection, and the use of de-identified data for research dissemination.

Informed consent

Informed consent to participate was obtained from all participants through an online information and consent form presented at the start of the survey. Consent was provided electronically by each participating undergraduate student before any responses were recorded. Data collection took place during the ethics-approved period from October 16, 2024 to October 25, 2024. The consent covered voluntary participation, the use of de-identified data for analysis and publication, and the right to withdraw at any time without penalty. Participants were informed about the study purpose, the measures taken to protect anonymity, how data would be stored and used, and that the study involved no more than minimal risk.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1057/s41599-026-06525-4.

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